

## SOFTWARE TOOL ARTICLE

# CADEMAS-ML: A Software Tool for Context-Aware Cooperative and Explainable Automated Decision-Making

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### ABSTRACT

Complex real-world decision problems often involve multiple decision-making actors that must combine their assessments while accounting for the organizational, operational, and strategic context surrounding the problem. The Cooperative Automated Decision-Making Systems (CADEMAS) framework was recently proposed to represent this form of cooperative and context-aware decision-making. However, previous CADEMAs instantiations cover only specific configurations of a framework that allows substantially broader combinations of automated decision components, contextual models, and aggregation mechanisms. This flexibility, while valuable for practical applications, also makes the framework difficult to operationalize without dedicated software support. This paper presents CADEMAs-ML, a software tool that implements a configurable class of CADEMAs systems in which automated decision-making components are machine learning models and contextual knowledge is represented through fuzzy logic. The tool supports the integration and weighting of multiple predictive models, configurable fuzzy context models, alternative aggregation operators, and an adjustable trade-off between predictive risk and contextual suitability. CADEMAs-ML also incorporates explainability and robustness analysis to support the inspection and assessment of resulting decisions. An employee attrition case study illustrates how the same predictive evidence can lead to different prioritizations under alternative contextual scenarios.

### KEYWORDS

automated decision-making systems; fuzzy decision making; context-aware decision support; machine learning; prioritization; cooperative systems

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## 1. Introduction

Decision-making in real-world organizations is rarely a matter of applying a single analytical model to a well-defined set of inputs. Complex decisions often involve multiple stakeholders, organizational units, or analytical components that contribute different pieces of evidence to the same problem. These contributions may reflect different objectives, information sources, areas of expertise, or levels of reliability, and therefore need to be coordinated before they can support a coherent course of action.

This perspective becomes particularly important when automated decision-making is used to support prediction, classification, or risk assessment. An automated decision-making (ADM) component can produce a probability, rating, recommendation, or risk score, but such an output does not by itself determine how the result should be acted upon. The appropriate course of action may depend on factors that are external to the predictive model, including organizational priorities, available resources, departmental objectives, regulatory constraints, or the strategic conditions under which the decision is made. In other words, decisions are taken in a “context” and not in vacuum.

Besides this, the simultaneous use of different ADM components may provide complementary views of the cases under consideration, while the final decision may also need to reflect contextual knowledge provided by domain experts or organizational policies.

These considerations motivated the development of the Cooperative Automated Decision-Making Systems (CADEMAS) framework proposed by Novoa-Hernández, Pelta, Godz, Verdegay, and Buendia-Carrillo (2026). CADEMAs provides a conceptual framework for coordinating heterogeneous automated decision-making components while explicitly incorporating the context in which their outputs are interpreted. Its central principle is to distinguish between the generation and integration of decision evidence and the contextual evaluation of that evidence. Multiple automated decision units can therefore contribute to a collective assessment, while a separate contextual layer represents organizational priorities, policies, or scenario-dependent considerations. The resulting decision is obtained by combining these two sources of information rather than treating the predictive assessment as an isolated or definitive outcome.

The work is inspired in the development of cooperative and multi-agent decision-support systems, in which heterogeneous decision components can contribute to a common decision process (Bui & Lee, 1999; Vahidov & Fazlollahi, 2004) and also, in multiperson and group decision-support research, where heterogeneous sources of information and preferences have traditionally been combined through aggregation, negotiation, and consensus-seeking mechanisms (Herrera-Viedma, Herrera, & Chiclana, 2002; Salo, 1995). Similar principles can be extended to automated settings, where the cooperating components could be software-based decision units rather than human decision makers.

A particular scenario where a CADEMAs-like tool is relevant is the following: suppose that you have a set of entities and some limited intervention budget. The question is how to prioritize the entities for receiving an intervention?

Examples of this scenario are attrition risk prevention Godz, Novoa-Hernández, and Pelta (2026), small and medium size companies financial rescue Navarro-Galera, Lara-Rubio, Novoa-Hernández, and Cruz Corona (2025), students dropout prevention (BUSCAR REF), and so on.

Using CADEMAs in these problems means that first, a risk is assessed for every entity using different ADMs (implementing machine learning techniques) and second, the suitability of the entity to a given context is also calculated. For example, in a context of economic crisis you may prioritize a company with a high number of employees (or with the lower); in a context of growing, you may prioritize companies with a high expected impact. Such contexts can be modeled, for example, with fuzzy rules as proposed in (Lamata, Pelta, & Verdegay, 2020; Pérez-Cañedo, Porras, Pelta, & Verdegay, 2023), allowing to combine quantitative information with expert knowledge and imprecise or linguistic assessments.

But also, users may need to understand why a particular entity receives a given prediction and how changes in the underlying information affect the result. Explainable artificial intelligence has consequently become an important area of research for the practical use of machine-learning models, particularly when predictions contribute to decisions with organizational or human consequences (Barredo Arrieta et al., 2020). Model-agnostic approaches such as SHAP provide local feature attributions that can help users inspect the factors contributing to individual predictions (Lundberg et al., 2020; Lundberg & Lee, 2017).

Robustness analysis is another required capability for context-aware decision support. In CADEMAs, the final prioritization depends not only on predictive evidence but also on contextual definitions, aggregation mechanisms, and the relative importance assigned to predictive and contextual information. Changes in these elements may alter the ordering of cases even when the underlying predictive models remain unchanged. Allowing users to examine such changes can provide information about the stability of the resulting decisions and expose cases whose prioritization is particularly sensitive to the adopted policy configuration (Barredo Arrieta et al., 2020).

To the best of our knowledge, there is no integrated tool that allows to deal with prioritization problems in the way described before.

In this scenario, the main aim is to present CADEMAs-ML, a software tool designed to engineer a configurable class of CADEMAs systems (to deal with prioritization problems) in which automated decision-making components are implemented as machine-learning models and contextual knowledge is represented through fuzzy rules. It provides configurable mechanisms for representing and aggregating contextual conditions, and combines predictive risk with contextual suitability through an explicit

trade-off mechanism. In addition to these core CADEMAs capabilities, this work proposes explainability and robustness analysis as natural extensions of the framework. Both capabilities are incorporated into CADEMAs-ML.

The remainder of the paper is organized as follows. Section 2 reviews the CADEMAs framework and related work on cooperative and fuzzy decision-support systems. Section ?? presents the architecture and main components of CADEMAs-ML. Section ?? reports the outputs produced by its modules. Both sections use an employee-attrition case study to illustrate the functionality and operation of the tool. Finally, Section 5 summarizes the main findings and discusses the implications of the proposed tool and possible directions for future work.

## 2. Background and related works

This section sets out the conceptual foundations on which CADEMAs-ML builds. We first revisit the CADEMAs framework in its general form (Section 2.1). We then situate the tool among related cooperative-decision frameworks and application-oriented decision-support systems, clarifying the gap it is designed to fill (Section 2.2).

### 2.1. CADEMAs framework

As introduced in Section 1, CADEMAs (Novoa-Hernández et al., 2026) provides a general conceptual and architectural framework for structuring cooperation among heterogeneous Automated Decision-Making (ADM) units. Its scope is not restricted to prediction. Organizational decisions may involve several decision-producing components, including classifiers, optimization models, simulation components, rule-based systems, large language model agents, or institutional procedures. These components may differ in their assumptions, data access, output types, and reliability. Rather than prescribing a particular learning method, rule formalism, or coordination protocol, CADEMAs defines how heterogeneous decision-producing components can contribute to a common decision process and how their outputs can subsequently be considered in the context of the problem.

A central premise of CADEMAs is that an ADM unit does not necessarily produce a complete operational decision. Instead, it may provide a *partial decision signal*, such as a score, prediction, rule activation, explanation, or suggested action. The semantics of these signals may also differ across units. They therefore need to be integrated before a final decision is obtained and may subsequently be adjusted according to the relevant context. Formally, a CADEMAs instance is represented by the tuple

$$\mathcal{S} = \langle \mathcal{IM}, \mathcal{A}, \mathcal{OM}, \mathcal{C}, \mathcal{T} \rangle, \quad (1)$$

where  $\mathcal{IM}$  is the Input Manager,  $\mathcal{A}$  is the set of ADM units,  $\mathcal{OM}$  is the Decision Integration Layer,  $\mathcal{C}$  is the Contextual Modulator, and  $\mathcal{T}$  is the universe of admissible data types and representations.

The Input Manager  $\mathcal{IM}$  routes, transforms, and validates the information entering the architecture. It distinguishes training or calibration data, which are used offline to construct or update the internal models of the ADM units, from decision-case data provided when a new case is evaluated. The set of ADM units is denoted by

$$\mathcal{A} = \{ADM_1, \dots, ADM_n\}. \quad (2)$$

Each  $ADM_i$  receives a data representation  $d_i \in \mathcal{T}$  and produces an output  $o_i \in \mathcal{T}$ . Cooperation among these units may be *indirect*, with each unit contributing an independent signal, or *direct*, with units exchanging intermediate representations. CADEMAs allows for both forms without imposing a specific communication protocol.

Given the heterogeneous collection  $\mathcal{O} = \{o_1, \dots, o_n\}$ , the Decision Integration Layer  $\mathcal{OM}$  synthesizes a unified decision object  $D$  through a mapping

$$\Phi : \mathcal{O} \rightarrow D.$$

This layer addresses the harmonization of ADM outputs and their subsequent aggregation or synthesis. The resulting  $D$  is a *raw* decision proposal: the outputs have been brought into a common decision representation, but the proposal has not yet been adjusted to the context in which it will be used. The Contextual Modulator provides this additional step by modifying the integrated proposal according to the relevant contextual conditions. Formally,

$$M : D \times \mathcal{C} \rightarrow D, \quad D^{\text{final}} = M(D^{\text{raw}}, \mathcal{C}), \quad (3)$$

where  $\mathcal{C}$  may encode regulatory requirements, organizational policies, resource constraints, or other scenario-dependent conditions. The framework does not prescribe a single representation for contextual information. Depending on the application, it may be expressed through crisp rules, utility functions, regulatory constraints, fuzzy propositions, or other complementary mechanisms.

The predictive and fuzzy specialization implemented by CADEMAs-ML, including cooperative risk scores, fuzzy context alignment, and an explicit trade-off between these components, is described in Section ???. The general CADEMAs architecture remains open with respect to the type of ADM units, the semantics of their outputs, the harmonization and aggregation mechanisms implemented by  $\Phi$ , and the way contextual information is incorporated through  $M$ .

This flexibility is reflected in the applications reported to date. The original for-

mulation illustrated the framework with an SME insolvency-prediction scenario in which heterogeneous decision units were combined with contextual constraints (Novoa-Hernández et al., 2026). Godz et al. (2026) subsequently instantiated CADEMAs for employee-attribution prevention, treating departmental machine-learning models as co-operating ADM units and encoding organizational scenarios through a fuzzy context layer. Building on this machine-learning pathway, Novoa-Hernández, Godz, and Pelta (2026) examined the effect of different aggregation operators on prioritization, comparing linear, weighted-geometric, and minimum operators and analyzing how their aggregation behavior affects the interaction between predictive evidence and contextual requirements. More recently, Vazquez-Sánchez, Buendía-Carrillo, and Corona (2026) proposed a cooperative ADM strategy with fuzzy contextual modulation for SME insolvency prediction, combining heterogeneous survival models, a contextual modulator, and explainable-AI support.

These studies illustrate different ways in which the CADEMAs architecture can be instantiated, but each requires a specific implementation for the domain and configuration considered. A reusable software environment is therefore useful for constructing, inspecting, comparing, and reproducing CADEMAs configurations. CADEMAs-ML addresses this need by providing such an environment for a machine-learning-based specialization of the framework.

## ***2.2. Related frameworks and decision-support applications***

Recent research has increasingly recognized that effective decision support requires more than accurate isolated models. In many application domains, decisions emerge from the interaction between automated units, expert knowledge, and contextual considerations that may change over time. For clarity, it is useful to distinguish two related but different bodies of work: general cooperative-decision frameworks, which address how multiple decision entities interact, and application-oriented decision-support systems, which implement concrete methods for a domain or family of decision problems.

At the framework level, cooperative game theory offers one formal perspective on how decision-making entities with heterogeneous roles or attributes may combine their contributions, including settings in which each agent is characterized by multiple attributes rather than a single payoff value (Borkotokey, Hazarika, & Mesiar, 2019). CADEMAs does not adopt this game-theoretic formulation, but shares its concern with making cooperation among heterogeneous entities explicit. At the operational level, Tariq, Chhetri, Nepal, and Paris (2024) propose an AI-to-collaborator (A2C) framework in which humans and AI systems share decision authority through explicit interaction mechanisms. Related work on hybrid intelligence emphasizes that authority distribution and human interpretive capacity should be modeled separately, because the level of AI integration and the cognitive flexibility of human decision-makers jointly

shape decision quality and accountability (Deliu & Moldovan, 2026). A neighboring line of work examines cooperation among multiple AI agents, including LLM-based multi-agent reinforcement learning (Sun, Huang, & Pompili, 2024). These contributions provide relevant points of comparison because they address cooperation among decision entities. However, their main focus is not the reusable implementation of a CADEMAs-like workflow in which heterogeneous ADM outputs are explicitly separated from context modulation and final prioritization.

A related theoretical thread connects cooperative game theory to case-level explanation. The Shapley value distributes a joint outcome among players according to their averaged marginal contributions (Borkotokey et al., 2019) and has been adopted in explainable artificial intelligence as a principled attribution mechanism. SHapley Additive exPlanations (SHAP) (Lundberg & Lee, 2017) treat model features as players and decompose each prediction into additive feature contributions with axiomatic guarantees; recent surveys document the breadth of this cooperative-to-XAI bridge (Li, Sun, Huang, & Chen, 2024). This literature motivates the inclusion of an explicit explanation module in CADEMAs-ML. The current implementation does not compute Shapley values over features: because the uploaded H2O MOJO models are treated as black boxes and may include stacked ensembles, the tool uses a model-agnostic perturbation procedure instead. The purpose is nonetheless the same as in Shapley-inspired XAI: to make the local drivers of a recommendation inspectable alongside the cooperative and contextual scores.

At the application and decision-support level, substantial progress has been made in fuzzy decision support, multi-criteria decision making, and hybrid fuzzy-ML workflows. Applications such as MAKEDECISION (Więckowski & Sałabun, 2024) and the fuzzy DEMATEL-based tool of Chekry et al. (2024) emphasize transparency and traceability by exposing intermediate calculations to users. A related stream has explored hybrid pipelines that combine expert judgment, causal weighting, and predictive components in domain-specific settings (Abdulgader, Eid, & Daneshvar Rouyendegh, 2018; Özkurt, Canay, Tunç, Aydın, & Velioğlu, 2025; Sahraei et al., 2022). Recent work has also extended the fuzzy toolbox through aggregation operators that account for interactions among decision attributes, capturing dependency structures that simple weighted averages cannot represent (Kacprzyk, Sirbiladze, & Tsulaia, 2022). The importance of explainability in AI-based decision support has been broadly recognized: any automated system whose outputs must be justified or audited requires transparent mechanisms to convey its reasoning to stakeholders (Barredo Arrieta et al., 2020), a requirement increasingly addressed through XAI techniques, including Shapley-value attribution (Li et al., 2024; Lundberg & Lee, 2017), in hybrid ranking and prediction pipelines (Özkurt et al., 2025). These contributions demonstrate the value of combining data-driven evidence with contextual or expert-defined criteria, but most are tailored to particular methods or domains rather than organized around the CADEMAs de-

composition of ADM units, integration, context, and final prioritization.

Open-source fuzzy and MCDM libraries, including PYFDM (Więckowski, Kizielewicz, & Sałabun, 2022), PYIFDM (Więckowski, Kizielewicz, & Sałabun, 2023), the IFMCDM package of Roszkowska, Jefmański, Dudek, and Kusterka-Jefmańska (2024), the ELIGERE platform (Grazioso, Caporaso, & di Gironimo, 2023), and the intuitionistic fuzzy dimensional-analysis approach of Pérez-Domínguez, Callejas-Cuervo, and García-Luna (2024), are also relevant. They improve reproducibility and provide reusable computational building blocks.

Table 1 summarizes representative application-oriented contributions in these areas. Three observations emerge from this comparison. First, published systems already provide strong ingredients for transparent decision support, including human–AI collaboration, auditable fuzzy MCDM, hybrid ranking pipelines, and reusable software libraries (Özkurt et al., 2025; Tariq et al., 2024; Więckowski et al., 2022; Więckowski & Sałabun, 2024). Second, these ingredients are usually developed in isolation. Third, recent CADEMAs applications already combine cooperative multi-ADM integration with an explicit contextual layer (Godz et al., 2026; Novoa-Hernández et al., 2026; Vazquez-Sánchez et al., 2026), yet they remain domain-specific configurations rather than a reusable software workflow with case-level explanation. What remains missing is an operational implementation that keeps these capabilities configurable and auditable together.

CADEMAS-ML is designed to address this gap for the predictive-model setting. Building on the framework and its attrition instantiation discussed above, it provides a configurable software implementation in which ML-based ADM outputs, contextual reasoning, and case-level explanation coexist as explicit and auditable components of the decision-making process. The source code is publicly available under an MIT license (Section 5); the main contribution is the operationalization of the CADEMAs decomposition in a reusable decision-support tool.

### 3. Instantiation of CADEMAs for the Prioritization Problem

#### 3.1. *Problem definition and overall mapping*

As previously noted in Section 1, prioritization problems in organizational settings arise when a set of cases must be ordered according to their relative need for attention, intervention, or resource allocation. Examples include identifying employees who may require retention measures, allocating limited resources among clients or cases, and scheduling inspections according to risk and organizational priorities. Let  $\mathcal{X} = \{x_1, \dots, x_N\}$  denote the set of cases considered in a decision process, where each case  $x_i$  is characterized by a set of observable attributes. The objective is not only to assign a score to each case, but to establish a priority ordering over  $\mathcal{X}$ .

**Table 1.** Comparison of representative application-oriented decision-support works related to CADEMAs-ML.

| Work                          | Decision-support focus                                                                                   | Multiple ADM units | Explicit context layer | Case-level XAI | Relation to CADEMAs-ML                                                                           |
|-------------------------------|----------------------------------------------------------------------------------------------------------|--------------------|------------------------|----------------|--------------------------------------------------------------------------------------------------|
| Tariq et al. (2024)           | Human–AI collaborative decision framework with automated, augmented, and collaborative modes             | Partly             | Partly                 | No             | Strong framework-level precedent, but not a configurable fuzzy/ML prioritization workflow        |
| Więckowski and Sałabun (2024) | Auditable fuzzy MCDM through a web-based tool                                                            | No                 | Partly                 | No             | Strong transparency, but no cooperative ADM integration layer                                    |
| Chekry et al. (2024)          | Open-source fuzzy DEMATEL decision-support tool                                                          | No                 | Partly                 | No             | Traceable fuzzy causal analysis, but method-specific rather than CADEMAs-like                    |
| Özkurt et al. (2025)          | Hybrid fuzzy MCDM, ML, and XAI for transparent ranking and prediction                                    | Partly             | Partly                 | Partly         | Combines ranking, prediction, and XAI, but lacks a general cooperative ADM architecture          |
| Sahraei et al. (2022)         | Fuzzy DEMATEL/BWM/ANP pipeline for spatial risk prioritization                                           | No                 | Partly                 | No             | Explicit causal weighting under uncertainty, but application-specific                            |
| Godz et al. (2026)            | Employee-attribution prevention using CADEMAs with departmental ML models                                | Yes                | Yes                    | No             | Direct CADEMAs application that motivates the implemented workflow                               |
| Novoa-Hernández et al. (2026) | Aggregation semantics for CADEMAs prioritization under alternative operators                             | Yes                | Yes                    | No             | Analyzes how linear, geometric, and minimum rules preserve or compensate contextual requirements |
| Vazquez-Sánchez et al. (2026) | Cooperative ADM with fuzzy contextual modulation for SME insolvency prediction                           | Yes                | Yes                    | Yes            | Extends CADEMAs-style cooperation to survival models with contextual modulation and XAI support  |
| CADEMAs-ML (this work)        | Configurable prioritization with ML-based ADM units, fuzzy context scenarios, and local case explanation | Yes                | Yes                    | Yes            | Operationalizes CADEMAs for reusable, explainable ML-driven decision support                     |

Let

$$P : \mathcal{X} \rightarrow [0, 1] \quad (4)$$

denote a priority function that assigns a normalized priority score to each case. This

score induces a preference relation over the cases such that

$$x_i \succeq_P x_j \iff P(x_i) \geq P(x_j). \quad (5)$$

The corresponding ranking can be represented by a function

$$\pi_P : \mathcal{X} \rightarrow \{1, \dots, N\}, \quad (6)$$

where lower values indicate higher priority. In the absence of ties, the rank of a case is therefore

$$\pi_P(x_i) = 1 + |\{x_j \in \mathcal{X} : P(x_j) > P(x_i)\}|. \quad (7)$$

Thus, a prioritization configuration defines both a priority score for each case and an ordering of the cases induced by those scores.

In practical settings, however, the priority function may need to reflect more than a single source of evidence. Different predictive or decision-making components may capture complementary aspects of a case, while the relevance of their outputs may depend on the organizational, operational, or regulatory context in which the prioritization is performed. A case with a high predicted risk, for example, may not have the same priority under different organizational objectives or contextual conditions. The prioritization problem can therefore be viewed as the combination of multiple partial assessments with contextual considerations before the final ordering is established.

This structure makes prioritization a natural application of the CADEMAs framework introduced in Section 2.1. CADEMAs provides an architecture in which heterogeneous Automated Decision-Making (ADM) units can contribute partial decision signals, these signals can be integrated into a common decision representation, and the resulting proposal can subsequently be considered in relation to the relevant context. CADEMAs-ML specializes this architecture for the prioritization problem by implementing the ADM units as machine-learning models and providing explicit mechanisms for defining the data, model, and contextual components involved in the decision process. The resulting specialization can be summarized through the mapping

$$\mathcal{S} = \langle \mathcal{IM}, \mathcal{A}, \mathcal{OM}, \mathcal{C}, \mathcal{T} \rangle$$

as follows:

- **Data representation ( $\mathcal{T}$ ):** The admissible data representations are specialized to the tabular case data and the decision signals required by the predictive and contextual components. This provides a common data basis while allowing indi-

vidual ADM units and contextual models to use only the variables relevant to them.

- **Input Manager ( $\mathcal{IM}$ ):** The Input Manager coordinates the information required by the different components of the system. In particular, it defines how case data are provided to the ADM units, which features are required by each predictive model, and which variables are available for contextual modeling. These requirements are specified independently of the execution of the models, allowing different predictive models, their reported performance measures, and contextual definitions to be incorporated without modifying the underlying decision process.
- **ADM units ( $\mathcal{A}$ ):** The ADM layer is realized through a collection of pre-trained machine-learning models that represent different predictive perspectives on the cases. Each model may use a different subset of the available attributes and contributes its own assessment of the target event. Their independent predictions constitute the partial decision signals that will subsequently be integrated.
- **Decision Integration Layer ( $\mathcal{OM}$ ):** The integration layer combines the predictive assessments provided by the ADM units into a common representation of predictive risk. The contribution of each model can be determined from its reported predictive performance, allowing models with different predictive quality to participate in the cooperative decision without requiring them to be retrained or structurally modified.
- **Contextual Modulator ( $\mathcal{C}$ ):** The contextual component represents the conditions under which the predictive evidence is interpreted. In CADEMAs-ML, contextual definitions specify which attributes of the input cases are relevant to a particular decision scenario and how their values contribute to contextual suitability. The resulting contextual assessment is then incorporated into the prioritization together with the integrated predictive evidence.

The following subsections specify the realization of the predictive integration and contextual modulation, including the mechanisms used to configure and combine their respective contributions. The final subsection introduces the explainability and robustness extensions added to this specialization.

### 3.2. *Input Manager, ADM Units and Data Representation*

In CADEMAs-ML, the Input Manager  $\mathcal{IM}$  materializes the information required to configure and execute the different components of the prioritization process. It is represented as the tuple

$$\mathcal{IM} = \langle \mathcal{M}, \mathcal{M}_{def}, \mathcal{C}_{def}, \mathcal{X} \rangle, \quad (8)$$

where:

- $\mathcal{M} = M_1, \dots, M_K$  is the collection of pre-trained machine-learning models used as ADM units. Each model provides a partial predictive assessment of the cases and may rely on a different subset of the available features.
- $\mathcal{M}_{def} = m_1, \dots, m_K$  contains the corresponding model definitions. Each definition specifies the metadata required to use a model within the CADEMAs instance, including its identifying information, predictive performance measures, and the features required as input.
- $\mathcal{C}_{def}$  is the collection of contextual definitions used to configure the contextual component of the instance. A contextual definition specifies the case attributes considered relevant to a particular scenario and the corresponding mechanisms used to evaluate their contextual suitability.
- $\mathcal{X} = x_1, \dots, x_N$  is the set of cases to be prioritized. Each case is represented by a set of observable features sufficient to satisfy the input requirements of the configured models and contextual definitions, i.e.,

$$f(x_i) \supseteq \bigcup_{k=1}^K f_k \cup f_C, \quad (9)$$

where  $f_k$  denotes the features required by model  $M_k$  and  $f_C$  the features required by the configured contextual definition.

The data representation  $\mathcal{T}$  is specialized to tabular case data and the structured representations required to configure models and contextual definitions. In particular, the case data provide the feature values consumed by the ADM units and contextual component, while model and context definitions provide the metadata and declarative specifications required for their execution.

It is worth noting that although these elements are defined as sets to support the evaluation of different configurations or scenarios, each CADEMAs-ML prioritization process operates on a specific set of models together with their metadata definitions, a single contextual definition, and a single set of case data.

Thus,  $\mathcal{IM}$  provides the data and configuration required by the subsequent CADEMAs components, with the main requirement that all features required by the configured models and contextual definition are present in the case data and mutually consistent. The corresponding file formats and examples are described in Section ?? and in the Supplementary Material.

### 3.3. Decision Integration Layer

The Decision Integration Layer  $\mathcal{OM}$  materializes the cooperation among the machine-learning-based ADM units by synthesizing their individual predictive outputs into a common decision representation. For a given prioritization scenario, each model  $M_k \in \mathcal{M}$  produces a prediction

$$p_{ik} = M_k(x_i), \quad p_{ik} \in [0, 1], \quad (10)$$

for case  $x_i$ . These individual predictions constitute the partial decision signals provided by the ADM units.

To account for differences in predictive performance, each model is assigned a non-negative weight  $w_k$ , with

$$w_k \geq 0, \quad \sum_{k=1}^K w_k = 1. \quad (11)$$

The integrated decision signal for case  $x_i$  is then obtained as

$$D_i^{\text{raw}} = \Phi(p_{i1}, \dots, p_{iK}) = \sum_{k=1}^K w_k p_{ik}, \quad (12)$$

where  $D_i^{\text{raw}} \in [0, 1]$  is the raw decision proposal resulting from the integration of the ADM outputs. In the present prioritization application, this value can be interpreted as an integrated *predictive score*, whose subsequent interpretation and combination with contextual information are determined by the Contextual Modulator.

The weighting mechanism implemented by CADEMAs-ML is based on the reported predictive performance of the models and is described below. The integration operates on the outputs of the ADM units without modifying their internal models, allowing heterogeneous predictive models to contribute to a common decision representation.

Although the resulting synthesis can technically be regarded as an ensemble-based prediction, CADEMAs-ML does not require the ADM units to have been trained jointly or with a common objective aimed at maximizing collective predictive performance. The models may therefore have been developed independently, using different training datasets, feature subsets, algorithms, or predictive objectives. Their cooperation takes place at the decision level, through the integration of their outputs within the CADEMAs architecture.

### 3.4. Contextual Modulator

The Contextual Modulator  $\mathcal{C}$  materializes the contextual component of CADEMAs-ML through a declarative specification of fuzzy contextual conditions. Following the role of context in Novoa-Hernández et al. (2026) and the use of fuzzy propositions in the attrition application of Godz et al. (2026), each context definition specifies the case attributes to be considered, the corresponding fuzzy propositions, and the aggregation rules used to obtain a contextual assessment. Unlike those application-specific implementations, CADEMAs-ML represents these definitions independently of the application code, allowing different contexts and decision policies to be configured through JSON specifications.

Formally, let  $V_r$  denote a case attribute and  $\tilde{F}_r$  a fuzzy set representing a linguistic predicate over the universe of  $V_r$ . For case  $x_i$ , an *atomic contextual rule* evaluates the proposition “ $V_r$  is  $\tilde{F}_r$ ” as

$$\mu_r(x_i) = \mu_{\tilde{F}_r}(x_{ir}) \in [0, 1], \quad (13)$$

where  $x_{ir}$  is the value of  $V_r$  for case  $x_i$ . CADEMAs-ML supports triangular, trapezoidal, linear increasing, and linear decreasing membership functions, as well as categorical maps and categorical sets.

Atomic rules may also include activation conditions. Let  $m_r(x_i) \in \{0, 1\}$  indicate whether the **when** condition of rule  $r$  is satisfied for case  $x_i$ . Its effective membership is then

$$\mu_r^*(x_i) = m_r(x_i)\mu_r(x_i), \quad (14)$$

with  $m_r(x_i) = 1$  for unconditional rules. This allows conditional policies to be expressed declaratively, without modifying the case data or introducing procedural logic.

Context definitions may further include *derived rules* that combine previously evaluated memberships. For a derived rule  $d$ , its membership is given by

$$\mu_d(x_i) = G_d \left( \mu_{r_1}^*(x_i), \dots, \mu_{r_q}^*(x_i) \right), \quad (15)$$

where  $G_d$  is an aggregation operator. The implemented operators include minimum-based **AND**, maximum-based **OR**, algebraic **PRODUCT**, and arithmetic **AVERAGE**. The contextual alignment score for case  $x_i$  is finally obtained as

$$C_i = H(\mu_1(x_i), \dots, \mu_L(x_i)), \quad (16)$$

where  $H$  is the final aggregation policy specified by the context definition.

The resulting  $C_i$  represents the degree to which case  $x_i$  satisfies the contextual conditions of the selected scenario. All atomic and derived memberships are retained by the tool, making the contextual assessment traceable to the underlying fuzzy rules rather than treating it as an opaque modification of the integrated predictive signal.

### 3.4.1. Contextual Modulation

The contextual assessment  $C_i$  is subsequently used by the Contextual Modulator to transform the raw decision proposal  $D_i^{\text{raw}}$  into the final decision value used for prioritization. In accordance with the general CADEMAs formulation, this modulation is not prescribed by the framework itself. CADEMAs-ML therefore implements three alternative aggregation operators for combining predictive and contextual evidence, namely the weighted arithmetic mean, the weighted geometric mean, and the minimum operator.

Let  $D_i^{\text{raw}} \in [0, 1]$  denote the integrated predictive signal and  $C_i \in [0, 1]$  the contextual alignment score. The final decision value is obtained as

$$D_i^{\text{final}} = M(D_i^{\text{raw}}, C_i). \quad (17)$$

For the weighted arithmetic operator, used in previous CADEMAs applications (Godz et al., 2026; Novoa-Hernández et al., 2026), the modulation is

$$D_i^{\text{final}} = \lambda D_i^{\text{raw}} + (1 - \lambda)C_i, \quad \lambda \in [0, 1]. \quad (18)$$

CADEMAs-ML additionally implements the weighted geometric operator,

$$D_i^{\text{final}} = (D_i^{\text{raw}})^\lambda (C_i)^{1-\lambda}, \quad (19)$$

and the minimum operator,

$$D_i^{\text{final}} = \min\{D_i^{\text{raw}}, C_i\}. \quad (20)$$

The inclusion of these alternatives follows the analysis in Novoa-Hernández et al. (2026), which examined the behavior of different aggregation operators for combining predictive and contextual signals. In particular, the geometric and minimum operators provide more restrictive forms of cooperation than the arithmetic operator, reducing the possibility that a low predictive signal or a low contextual alignment is compensated by a high value of the other component. The parameter  $\lambda$  controls the relative contribution of predictive and contextual information for the arithmetic and geometric operators, whereas the minimum operator has no weighting parameter.

### 3.5. *Explainability Extension*

CADEMAS-ML extends the original CADEMAs framework with a case-level explanation mechanism designed to make the cooperative decision process interpretable and auditable. While CADEMAs distinguishes the outputs of ADM units, their integration, and contextual modulation, it does not prescribe how these elements should be exposed to decision-makers. The proposed extension addresses this aspect by retaining intermediate computational artifacts and providing two complementary explanation layers: a predictive explanation, which identifies the local influence of the input features on the integrated predictive signal, and a contextual explanation, which traces the fuzzy evidence underlying the contextual assessment.

For a selected case  $x_i$ , the explanation preserves the aggregation mechanism used to obtain the final decision. The predictive and contextual component values,  $D_i^{\text{raw}}$  and  $C_i$ , are reported together with the selected aggregation operator and, when applicable, its parameter  $\lambda$ . For the linear operator, the two components can be interpreted additively. For the weighted geometric operator, the explanation reflects their multiplicative combination and the corresponding balance parameter. For the minimum operator, the lower component is identified as the limiting factor. This representation preserves the semantics of each aggregation mechanism without imposing an additive decomposition on operators for which such a decomposition is not defined.

To explain the predictive component, CADEMAs-ML employs a model-agnostic perturbation procedure suitable for the heterogeneous and potentially ensemble-based ML models supported by the tool. For each model  $M_k$  and feature  $f$ , the local effect is estimated by comparing the original prediction with the prediction obtained after replacing the observed feature value with a cohort reference value  $b_f$ :

$$\Delta_{ikf} = M_k(x_i) - M_k\left(x_i^{(f \leftarrow b_f)}\right), \quad (21)$$

where  $b_f$  is the median of the reference cohort for numerical features and the mode for categorical features. The model-specific effects are then integrated using the same

weights employed by the Decision Integration Layer:

$$\Delta_{if} = \sum_{k=1}^K w_k \Delta_{ikf}. \quad (22)$$

Positive values indicate that the observed feature value increases the integrated predictive score relative to the reference value, whereas negative values indicate a decrease. These quantities provide approximate local feature effects in probability points and are not interpreted as causal effects. The resulting predictive explanation can therefore be represented as

$$\mathcal{E}_i^{\text{pred}} = \{(f, \Delta_{if})\}_{f \in \mathcal{F}}, \quad (23)$$

where  $\mathcal{F}$  denotes the set of features considered in the perturbation analysis.

The contextual explanation traces the assessment  $C_i$  back to the fuzzy propositions and derived rules defined in the selected context. By retaining the membership values of the evaluated rules, the tool exposes the intermediate evidence underlying the contextual aggregation:

$$\mathcal{E}_i^{\text{context}} = \{(r, \mu_r^*(x_i))\}_{r \in \mathcal{R}_a} \cup \{(d, \mu_d(x_i))\}_{d \in \mathcal{R}_d}, \quad (24)$$

where  $\mathcal{R}_a$  and  $\mathcal{R}_d$  denote the sets of atomic and derived rules evaluated for case  $x_i$ , respectively. The use of  $\mu_r^*(x_i)$  for atomic rules accounts for possible activation conditions. This representation allows the contextual assessment to be inspected in terms of the conditions satisfied by the case, including conditional rule activations and the aggregation operations leading to  $C_i$ .

The two explanation layers are subsequently combined into a deterministic natural-language representation. Formally, the explanation for case  $x_i$  can be represented as

$$E_i = \Psi \left( D_i^{\text{final}}, D_i^{\text{raw}}, C_i, \mathcal{E}_i^{\text{pred}}, \mathcal{E}_i^{\text{context}}, \mathcal{A}_i \right), \quad (25)$$

where  $\mathcal{A}_i$  denotes the selected aggregation mechanism and its parameters when applicable. The function  $\Psi$  translates the quantitative evidence into a structured textual explanation that relates the final prioritization outcome to its predictive and contextual drivers. In this way, the natural-language layer does not replace the underlying explanation artifacts, but provides a more accessible interpretation of how the predictive

evidence, contextual suitability, and selected cooperation mechanism jointly determine the final decision.

### 3.6. *Robustness Extension*

CADEMAS-ML extends the CADEMAs decision process with a robustness analysis aimed at assessing the stability of prioritization results under alternative contextual modulation configurations. Let

$$\mathcal{G}_{\text{mod}} = \{g = (o, \theta_o)\}$$

denote the set of modulation configurations explored by the analyst, where  $o$  identifies the selected aggregation operator and  $\theta_o$  denotes its associated parameterization, which may be empty for parameter-free operators. Each configuration  $g \in \mathcal{G}_{\text{mod}}$  induces a prioritization score vector

$$\mathbf{D}^{(g)} = \left(D_1^{(g)}, \dots, D_N^{(g)}\right)$$

and an associated ranking  $\pi^{(g)}$  over the cases.

This representation allows robustness to be examined from two complementary perspectives. Within-operator robustness evaluates the stability of prioritization results under different parameterizations of the same aggregation operator. For operators with a balance parameter, such as the linear and weighted geometric operators, the analyst can explore a set of  $\lambda$  values and compare the resulting score vectors and rankings. The minimum operator provides a parameter-free configuration that can be included as a reference in the broader analysis.

Across-operator robustness evaluates the stability of prioritization results when the aggregation mechanism itself is changed. Given the same predictive and contextual signals, different operators in the set

$$\mathcal{O} = \{\text{linear, geometric, minimum}\}$$

can generate different score vectors and induced rankings. Comparing these configurations makes it possible to identify cases whose prioritization remains stable across alternative modulation mechanisms and cases whose scores or ranking positions are sensitive to the selected operator.

More generally, the tool considers the collection

$$\mathcal{R}_{\mathcal{G}} = \left\{ \pi^{(g)} : g \in \mathcal{G}_{\text{mod}} \right\}$$

of rankings induced by the explored configurations. This collection can be inspected either individually or simultaneously, allowing the analyst to examine changes in the relative ordering of cases, shifts in their prioritization scores, and changes in the cases occupying the highest-priority positions. The same framework can therefore be used to perform within-operator and across-operator comparisons independently or jointly.

The robustness analysis does not assume that one aggregation mechanism provides a universally preferred prioritization. Instead, it exposes the extent to which prioritization conclusions depend on the contextual modulation configuration. In particular, it allows the analyst to distinguish cases whose prioritization is stable across configurations from cases whose position is sensitive to changes in the balance between predictive evidence and contextual suitability or to the aggregation mechanism itself.

## 4. Architecture and main functionalities

### 4.1. *Software architecture*

### 4.2. *Main functionalities*

#### 4.2.1. *Demonstration setup: Employee-attribution case study*

#### 4.2.2. *Integrated prioritization overview*

#### 4.2.3. *Cooperative predictive risk audit*

#### 4.2.4. *Contextual alignment evaluation*

#### 4.2.5. *Case-level explanation*

#### 4.2.6. *Robustness and sensitivity analysis*

## 5. Discussion

This paper presented CADEMAs-ML as an executable specialization of the CADEMAs framework for cooperative, context-aware, ML-based prioritization. Its contribution is not a new attrition predictor, but an operational separation of predictive evidence, contextual evaluation, and final prioritization. This separation builds on the general architecture of Novoa-Hernández et al. (2026) and the employee-attribution instantiation of Godz et al. (2026), while adding reusable configuration, case-level explanation, and ranking-sensitivity analysis. The employee-attribution case study shows the practical consequence of that design: rankings based on cooperative predictive risk need not coincide with rankings based on contextual alignment, and the explicit combination of both can be inspected rather than treated as an opaque post-processing step.

The tool also occupies a distinct position relative to related decision-support software. Existing platforms and libraries have made important advances in exposing intermediate fuzzy or multi-criteria calculations (Chekry et al., 2024; Więckowski & Salabun, 2024), while hybrid approaches have combined prediction, ranking, expert knowledge, and explainability in application-specific pipelines (Özkurt et al., 2025). CADEMAs-ML complements these contributions by organizing the workflow around cooperating ADM units and by keeping their integrated output separate from the context in which a decision is made. The *Models*, *Context*, and *Overview* views preserve this distinction; the *Explain* and *Robustness* views then support case-level interpretation and policy-sensitivity analysis. These additions extend the static prioritization workflow considered in earlier CADEMAs work.

For practitioners, the main benefit of this architecture is the ability to revise decision policy without embedding that policy in the predictive models themselves. When organizational conditions change, analysts can retain the predictive layer, provide a different context specification, and examine how the resulting priorities change. The separate reporting of  $R_i$ ,  $C_i$ , and  $P_i(\lambda)$  makes the source of a recommendation visible, while the  $\lambda$  sweep identifies cases whose position is stable and cases whose priority depends strongly on the selected balance. This does not remove the need for professional judgment; rather, it provides a structured record of the predictive and policy assumptions presented to the decision-maker. Potential applications include alert triage, review queues, retention programs, and allocation of limited support or inspection resources, provided that models, contextual rules, and decision thresholds are validated for the domain concerned.

For researchers, CADEMAs-ML provides a reproducible environment in which alternative cooperation, context-representation, explanation, and robustness mechanisms can be compared within a common architecture. The current implementation uses performance-normalized weights to combine independently trained models and a convex linear rule,  $P_i(\lambda) = \lambda R_i + (1 - \lambda)C_i$ , to integrate predictive risk and context. This final rule is compensatory: sufficiently strong evidence in one component can offset weaker evidence in the other. Recent work on aggregation semantics for CADEMAs (Novoa-Hernández et al., 2026) motivates extending the software with operator families that express different compensation requirements, including conjunctive or veto-like behavior. Making the aggregation semantics configurable would allow the software to distinguish settings in which compensation is acceptable from those in which a minimum level of predictive or contextual support is required.

A related direction is to move cooperation upstream, from post-hoc output aggregation to model development. At present, the predictive ADM units are trained independently and cooperation begins when their probabilities are weighted. Future versions could investigate diversity-aware model selection, theory-informed learning constraints, or multi-objective procedures that balance predictive performance, cali-

bration, and complementarity. The aim would not be to encourage disagreement for its own sake, but to obtain models that remain individually reliable while contributing non-redundant evidence from different variables, data sources, or perspectives on the problem. Measures of conditional competence and redundancy could then inform both training and aggregation, providing a more principled trade-off between the performance of individual ADM units and the informational value of their cooperation.

Several limitations delimit these implications. The ADM units currently supported at run time are H2O MOJO classifiers; the broader CADEMAs notion of heterogeneous units, including optimization models, simulators, or rule-based systems, is not yet implemented. Local ML attributions are model-agnostic perturbations around a cohort baseline and should not be read as causal effects or as substitutes for model-specific explanation methods when model internals are available. The attrition case study uses a 20-case illustrative subset and is neither a predictive-performance study nor evidence for general human-resources policy. Moreover, context specifications require domain-expert elicitation, and the interface does not yet provide versioned approval workflows or immutable audit trails for production governance. Future development should therefore combine broader ADM support and richer aggregation semantics with empirical validation in additional domains, stronger context governance, and evaluation with the practitioners who would interpret and act on the resulting priorities.

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## **Disclosure statement**

The authors report there are no competing interests to declare.

## **Declaration of generative AI use**

Generative AI tools were used to assist with language editing and manuscript formatting. The authors reviewed and take full responsibility for the content of this article.

## **Data availability statement**

The illustrative employee records used in this article are a 20-case subset derived from the publicly available IBM HR Analytics Employee Attrition and Performance dataset

(Subhash, 2017). The subset, model metadata, MOJO archives, and context specifications are distributed with the software repository cited in Section 5.

### Software availability

CADEMAS-ML is available at <https://github.com/pnovo/cademas-app> under an MIT license (Novoa-Hernández & Pelta, 2026). An archival snapshot will be deposited in Zenodo at the time of submission; the corresponding DOI will be inserted when it is available.

### Supplementary material

Supplementary Material S1 accompanies this article and provides the complete specification of the four application inputs, field-level constraints, cross-file consistency requirements, and minimal configuration examples.

### Author contributions

Pavel Novoa-Hernández: conceptualization, methodology, software, validation, writing – original draft, visualization. David A. Pelta: conceptualization, methodology, supervision, writing – review and editing.

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## Appendix A. Attrition example dataset and model features

The 20-case attrition example distributed with CADEMAs-ML contains 36 columns: `Case_ID`, 34 predictor variables, and the `attrition` outcome label. The predictors cover demographic, professional, compensation, performance, and organizational attributes from the *IBM HR Analytics Employee Attrition and Performance* Kaggle dataset used in Godz et al. (2026). Five variables (`age`, `gender`, `department`, `job_role`, and `job_level`) are shared by all four MOJO models. Two additional columns (`employee_count` and `employee_number`) are retained in the CSV file but are not used by any model in the example bundle. Each MOJO receives the full case matrix and selects its required variables according to the schema stored in the model-definition JSON file.

Table A1 summarizes the feature assignment. The Human Resources model reads `over18` from the CSV file; the corresponding field is named `over_18` in the model metadata.

## Appendix B. Context specification for the *Digital / Technological Transformation* scenario

The example repository includes two context configurations, but the walkthrough focuses on *Digital / Technological Transformation*. This appendix reports the corresponding fuzzy specification in closed form so that every membership degree and the final context-alignment score  $C_i$  can be audited without inspecting application code. The rules instantiate the generic formalism of Equations (13)–(16); equivalent parameters are stored in the repository JSON file consumed by CADEMAs-ML<sup>1</sup>. They are illustrative policy encodings, not hard-coded assumptions of the tool.

### B.1. *Generic membership functions*

Each numerical atomic rule is evaluated through one of four parametric membership functions. For an input value  $x$ , the *linear increasing* function with breakpoints  $(a, b)$ ,

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<sup>1</sup>[https://github.com/pnovoa/cademas-app/blob/main/example\\_attrition/context/context\\_digital\\_transformation.json](https://github.com/pnovoa/cademas-app/blob/main/example_attrition/context/context_digital_transformation.json).

**Table A1.** Predictor variables in the attrition example and the MOJO models that use each one.  $M_1$ – $M_4$  denote Culture and Wellbeing, Finance, Operations and Development, and Human Resources, respectively.

| Predictor                  | $M_1$<br>Culture &<br>Wellbeing | $M_2$<br>Finance | $M_3$<br>Ops. &<br>Dev. | $M_4$<br>Human<br>Resources |
|----------------------------|---------------------------------|------------------|-------------------------|-----------------------------|
| age                        | ✓                               | ✓                | ✓                       | ✓                           |
| business_travel            | ✓                               |                  |                         |                             |
| daily_rate                 |                                 | ✓                |                         |                             |
| department                 | ✓                               | ✓                | ✓                       | ✓                           |
| distance_from_home         |                                 |                  |                         | ✓                           |
| education                  |                                 |                  |                         | ✓                           |
| education_field            |                                 |                  |                         | ✓                           |
| employee_count             |                                 |                  |                         |                             |
| employee_number            |                                 |                  |                         |                             |
| environment_satisfaction   | ✓                               |                  |                         |                             |
| gender                     | ✓                               | ✓                | ✓                       | ✓                           |
| hourly_rate                |                                 | ✓                |                         |                             |
| job_involvement            | ✓                               |                  |                         |                             |
| job_level                  | ✓                               | ✓                | ✓                       | ✓                           |
| job_role                   | ✓                               | ✓                | ✓                       | ✓                           |
| job_satisfaction           | ✓                               |                  |                         |                             |
| marital_status             |                                 |                  |                         | ✓                           |
| monthly_income             |                                 | ✓                |                         |                             |
| monthly_rate               |                                 | ✓                |                         |                             |
| num_companies_worked       |                                 |                  | ✓                       |                             |
| over18                     |                                 |                  |                         | ✓                           |
| over_time                  | ✓                               |                  |                         |                             |
| percent_salary_hike        |                                 | ✓                |                         |                             |
| performance_rating         |                                 |                  | ✓                       |                             |
| relationship_satisfaction  | ✓                               |                  |                         |                             |
| standard_hours             |                                 |                  |                         | ✓                           |
| stock_option_level         |                                 | ✓                |                         |                             |
| total_working_years        |                                 |                  | ✓                       |                             |
| training_times_last_year   |                                 |                  | ✓                       |                             |
| work_life_balance          | ✓                               |                  |                         |                             |
| years_at_company           |                                 |                  | ✓                       |                             |
| years_in_current_role      |                                 |                  | ✓                       |                             |
| years_since_last_promotion |                                 |                  | ✓                       |                             |
| years_with_curr_manager    |                                 |                  | ✓                       |                             |
| <b>Total features</b>      | 12                              | 11               | 13                      | 11                          |

$a < b$ , is

$$\mu_{a,b}^{\uparrow}(x) = \begin{cases} 0, & x \leq a, \\ \frac{x-a}{b-a}, & a < x < b, \\ 1, & x \geq b, \end{cases} \quad (\text{B1})$$

and the *linear decreasing* function with breakpoints  $(a, b)$  is its mirror image

$$\mu_{a,b}^{\downarrow}(x) = 1 - \mu_{a,b}^{\uparrow}(x) = \begin{cases} 1, & x \leq a, \\ \frac{b-x}{b-a}, & a < x < b, \\ 0, & x \geq b. \end{cases} \quad (\text{B2})$$

The *triangular* function with support  $(a, c)$  and peak  $b$ ,  $a < b < c$ , is

$$\text{tri}_{a,b,c}(x) = \begin{cases} \frac{x-a}{b-a}, & a < x \leq b, \\ \frac{c-x}{c-b}, & b < x < c, \\ 0, & \text{otherwise,} \end{cases} \quad (\text{B3})$$

and the *trapezoidal* function with feet  $(a, d)$  and shoulders  $(b, c)$ ,  $a < b \leq c < d$ , is

$$\Pi_{a,b,c,d}(x) = \begin{cases} \frac{x-a}{b-a}, & a < x < b, \\ 1, & b \leq x \leq c, \\ \frac{d-x}{d-c}, & c < x < d, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{B4})$$

Categorical attributes are handled by a direct map  $\kappa : \mathcal{S} \rightarrow [0, 1]$  that assigns a degree to each symbolic value  $s \in \mathcal{S}$ , with unspecified categories defaulting to 0.

## B.2. Atomic rules

Seven atomic rules evaluate recent training intensity, education level, company tenure, tenure in the current role, age in the investment window, accumulated experience, and role adaptability to technological change. For case  $x_i$ , the corresponding memberships are

$$\mu_{\text{train}}(x_i) = \mu_{1,6}^{\uparrow}(\text{training\_times\_last\_year}_i), \quad (\text{B5})$$

$$\mu_{\text{edu}}(x_i) = \mu_{2,5}^{\uparrow}(\text{education}_i), \quad (\text{B6})$$

$$\mu_{\text{yrsCo}}(x_i) = \text{tri}_{3,10,20}(\text{years\_at\_company}_i), \quad (\text{B7})$$

$$\mu_{\text{yrsRole}}(x_i) = \text{tri}_{1,6,11}(\text{years\_in\_current\_role}_i), \quad (\text{B8})$$

$$\mu_{\text{age}}(x_i) = \Pi_{22,28,50,57}(\text{age}_i), \quad (\text{B9})$$

$$\mu_{\text{exp}}(x_i) = \mu_{2,10}^{\uparrow}(\text{total\_working\_years}_i), \quad (\text{B10})$$

$$\mu_{\text{role}}(x_i) = \kappa_{\text{role}}(\text{job\_role}_i), \quad (\text{B11})$$

where the role-adaptability map assigns expert scores to each job role:

$$\kappa_{\text{role}}(s) = \begin{cases} 0.9, & s \in \{\text{Research Scientist, Research Director}\}, \\ 0.8, & s = \text{Manager}, \\ 0.7, & s \in \{\text{Manufacturing Director, Healthcare Representative}\}, \\ 0.6, & s = \text{Sales Executive}, \\ 0.5, & s = \text{Human Resources}, \\ 0.4, & s \in \{\text{Laboratory Technician, Sales Representative}\}. \end{cases} \quad (\text{B12})$$

Each expression can be read directly: for instance, Equation (B9) states that ages between 28 and 50 fully satisfy the investment window ( $\mu = 1$ ), the degree decays linearly toward the feet at 22 and 57, and ages outside  $[22, 57]$  are excluded.

### B.3. Derived rules and final aggregation

The aggregation operators of Equation (15), applied to a vector  $\mathbf{u} = (u_1, \dots, u_q)$ , are

$$\text{AND}(\mathbf{u}) = \min_j u_j, \quad \text{OR}(\mathbf{u}) = \max_j u_j, \quad \text{PROD}(\mathbf{u}) = \prod_{j=1}^q u_j, \quad \text{AVG}(\mathbf{u}) = \frac{1}{q} \sum_{j=1}^q u_j. \quad (\text{B13})$$

With these operators, three derived criteria combine the atomic memberships: *upskilling capacity* (training and education), *career scalability* (company and role tenure), and *investment horizon* (age window and experience). Formally,

$$\mu_{\text{upskill}}(x_i) = \text{AVG}(\mu_{\text{train}}(x_i), \mu_{\text{edu}}(x_i)) = \frac{1}{2}(\mu_{\text{train}}(x_i) + \mu_{\text{edu}}(x_i)), \quad (\text{B14})$$

$$\mu_{\text{career}}(x_i) = \text{PROD}(\mu_{\text{yrsCo}}(x_i), \mu_{\text{yrsRole}}(x_i)) = \mu_{\text{yrsCo}}(x_i) \mu_{\text{yrsRole}}(x_i), \quad (\text{B15})$$

$$\mu_{\text{invest}}(x_i) = \text{PROD}(\mu_{\text{age}}(x_i), \mu_{\text{exp}}(x_i)) = \mu_{\text{age}}(x_i) \mu_{\text{exp}}(x_i). \quad (\text{B16})$$

The choice of operator carries a clear semantics: AVG lets strong training compensate for weaker formal education in *upskilling capacity*, whereas the PROD operator in *career scalability* and *investment horizon* is conjunctive, so a near-zero membership in either input collapses the derived degree toward zero. The context JSON also identifies four high-level criteria—upskilling, role adaptability, career scalability, and investment horizon—that can be evaluated as a declarative logic tree when no aggregation override is supplied.

In the reported demonstration, the *average* operator is selected explicitly in the user interface. This setting takes precedence over the declarative logic tree and averages all

numeric atomic and derived memberships retained in the audit matrix. Let

$$\mathcal{S} = \{\text{train, edu, yrsCo, yrsRole, age, exp, role, upskill, career, invest}\}.$$

The context-alignment score used in the figures and numerical results is therefore

$$C_i = \frac{1}{|\mathcal{S}|} \sum_{r \in \mathcal{S}} \mu_r(x_i) = \frac{1}{10} \sum_{r \in \mathcal{S}} \mu_r(x_i). \quad (\text{B17})$$

Equation (B17) is the explicit instance of the generic policy in Equation (16) used for this run and defines the contextual evidence surfaced in the *Context* module (Figure ??) and traced at case level in the *Explain* module. Selecting *minimum (strict)* or *product* in the interface applies the corresponding operator to the same audit-matrix columns.

## Appendix C. Input files and consistency requirements

CADEMAS-ML requires four coordinated inputs. Table C1 summarizes their roles and principal constraints; Supplementary Material S1 gives the complete field-level specification and minimal examples. The specification distinguishes fields required for scoring from metadata used for display and explanation.

**Table C1.** Summary of the CADEMAs-ML input contract.

| Input                        | Required content                                                                                                                                                                                                      | Cross-file constraints                                                                                                                                                                                                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Case dataset (CSV)           | One row per case and columns required by the uploaded models and context rules. A <b>Case_ID</b> or <b>ID</b> column is optional; otherwise, identifiers are generated.                                               | Column names and data types must be compatible with the MOJO schemas. Every feature referenced by a context rule must be present. Identifier columns are excluded from inference.                                                  |
| Model metadata (JSON)        | One top-level object per model, containing a <b>performance</b> object. The bundled schema also records <b>department</b> and <b>features</b> ; the latter is needed for the perturbation-based explanation workflow. | Each top-level key must exactly match an uploaded MOJO filename. The metric selected for weighting must be available for every participating model and should be a finite non-negative value.                                      |
| Predictive models            | H2O MOJO archives in <b>.zip</b> format that return compatible positive-class probabilities.                                                                                                                          | Each archive must have a matching metadata entry. All models must use the same positive-class interpretation, although their predictor subsets may differ.                                                                         |
| Context specification (JSON) | A context name, atomic <b>rules</b> , optional <b>derived_rules</b> , and an optional final <b>logic</b> tree. Rules identify a feature, membership type, parameters, and optionally an activation condition.         | Rule names must be unique, references in derived or final rules must resolve, and parameter shapes must match the membership type. The aggregation operator selected in the interface overrides the final JSON logic for that run. |

The application checks filename matching and loads the selected metric at run time, but publication-quality deployments should validate the complete bundle before analysis. Such validation includes checking finite metric values, membership-parameter or-

dering, unresolved rule references, missing features, incompatible categorical values, and agreement on the positive class. Keeping these constraints explicit is important because the four files jointly define the executable decision model: none of them is fully interpretable in isolation.